

On the periodogram estimator of period from sparse, noisy timing data

Barry G. Quinn
Department of Statistics
Faculty of Science
Macquarie University
Sydney, NSW, 2109, Australia
email: barry.quinn@mq.edu.au

I. Vaughan L. Clarkson
School of Information Technology
& Electrical Engineering
The University of Queensland
Qld., 4072, Australia
email: v.clarkson@uq.edu.au

Robby McKilliam
Institute For Telecommunications Research
University of South Australia
Mawson Lakes
SA, 5095, Australia
email: robby.mckilliam@unisa.edu.au

Abstract—The problem discussed is that of estimating the period of a sequence of periodic events when the occurrence time measurements are noisy and sparse. The problem arises in signal processing applications such as baud estimation from zero-crossings in telecommunications and in pulse repetition interval estimation in electronic support measures. Estimation techniques have been based on periodogram maximisation [1][2], Euclidean algorithms [3][4][5], least squares line search [6], lattice line search [7], Gaussian maximum likelihood [8] and least squares [9]. Aside from [9], there has been no rigorous statistical analysis. In this paper, we show that the periodogram maximiser has excellent (theoretical) asymptotic statistical properties, illustrating them via simulation.

Index Terms—Period estimation, pulse repetition interval, nearest lattice point problem, modified least squares.

I. INTRODUCTION

We examine a statistical model in which occurrence times, or times of arrival (TOAs), are measured. The TOA measurements arise from a process that exhibits an underlying periodicity. However, the TOAs do not necessarily correspond to consecutive periods. Some, or perhaps many, of the periodic events are missing from the TOA record. Furthermore, the TOAs are measured with noise. The aim is to estimate the period of the process from this sparse, noisy record.

Equivalently, the process may be viewed as a periodic pulse train from which the TOAs of only a finite number of non-consecutive pulses are measured with noise. This model is common to a number of signal-processing applications, including pulse repetition interval (PRI) estimation in electronic support measures (ESM), beat tracking in music analysis, baud estimation from zero-crossings and hop-rate estimation from frequency-hopped spread spectrum signals in telecommunications; see [1-11] and references therein.

Fogel and Gavish [1][2] were the first to suggest the statistical model considered here. In their model, the TOA of the n th pulse, X_n , is assumed to satisfy

$$X_n = \alpha + \beta S_n + \varepsilon_n; n = 1, \dots, N, \quad (1)$$

where α is the phase, β the period, $\{S_n\}$ a set of integer indices of the observed, enumerated pulses and $\{\varepsilon_n\}$ Gaussian white noise. They proposed estimating β by performing a “spectral analysis” on a delta function representation of the

TOAs, or maximizing the periodogram for point processes developed earlier by Bartlett [12] for detecting periodicities in point processes.

Recognising that, in the noiseless case, the time-differences of arrival (TDOAs) are commensurate with the period, Casey and Sadler [3][4] and Clarkson et al. [5] proposed modified Euclidean algorithms to estimate the period. While Fogel and Gavish [1][2] had assumed the S_n to be (nuisance) parameters, Clarkson et al. [5] modeled the increments $S_n - S_{n-1}$ statistically, assuming them to be independent and geometrically distributed.

Sidiropoulos et al. [6] proposed a line-search approach for estimating the period. Their work was extended in [7-9]. The approach can be motivated by the observation that, if the indices are known, the estimation of phase and period reverts to classical simple linear regression. On the other hand, given a trial value of phase and period, the corresponding most likely indices can be calculated using a simple rounding operation. By testing a number of trial values of period along an interval, the so-called ‘line search’, the indices can be estimated from which, in turn, refined estimators of period and phase can be derived. The resulting estimator of period is that which yields the least residual sum of squares from the line search.

Through this connection with linear regression, Sidiropoulos et al. [6] compared the practical performance of various techniques with what they termed the ‘clairvoyant’ Cramér-Rao Lower Bound, that is, the Cramér-Rao Lower Bound assuming the S_n to be known, or, equivalently, the variance of the least squares estimator conditional on the S_n . Clarkson [7] showed that when a priori bounds are placed on α and β in (1) to ensure identifiability, maximum likelihood (ML) estimators exist. McKilliam and Clarkson [8] showed that the line-search approach can be adapted to guarantee the calculation, in polynomial time, of an ML estimator. The authors of the present paper further showed in [9] that the least-squares method underpinning these techniques can be adapted to exhibit excellent asymptotic properties.

Apart from [9], which can be regarded as a companion paper (albeit one that can be read independently), statistical analysis in previous work on sparse, noisy period estimation has been confined to the low-noise case, for quite specific distributional

assumptions and carried out via simulation. In this paper, as in [9], we drop most of these assumptions. We show that the periodogram estimator is strongly consistent and satisfies a central limit theorem. Some of the methods of proof have been suggested by those introduced in [11] and [9], where again minimal conditions were imposed on the noise processes.

In Section II, we fully describe our statistical model. In Section III, we briefly review the periodogram of a point process and demonstrate its utility in period estimation. In Section IV, we illustrate through numerical simulations the main asymptotic results. These are stated and (to the extent that space permits) proved in Section V.

II. THE MODEL

In this paper we shall assume that $\{X_n\}$ satisfies (1), with β_0 equal to the true value of β , subject to the conditions

- 1) $\beta_L < \beta_0 \leq 2\beta_L$;
- 2) Let $U_n = \varepsilon_n/\beta_0$, where β_0 is the true value of β . Then the U_n are continuous, independent and identically symmetrically distributed random variables with mean 0 and finite absolute moment of order $2 + \nu$, for some $\nu > 0$, common variance σ_U^2 and common characteristic function $\phi_U(t) = Ee^{jtU_n}$;
- 3) $S_n = \sum_{i=1}^n Y_i$, where the Y_i are independent and identically distributed and discrete, with mean μ_Y , variance σ_Y^2 , and common characteristic function $\phi_Y(t) = Ee^{jtY_i}$ having the property that $\phi_Y(t) = 1$ only if t is a multiple of 2π .

Note that $\{\varepsilon_n\}$ is *not* assumed to be Gaussian, nor $\{Y_n\}$ geometric. Discussion of the above conditions is in order. Firstly, if β were unconstrained, we could replace β and S_n with β/k and kS_n for any integer k , and (1) would not change, since $\{kS_n\}$ would still be an integer-valued sequence. The third condition imposes minimal distributional assumptions on the times between observations. The condition on $\phi_Y(t)$ precludes, for example, the possibility that Y takes on only even values.

III. THE POINT PROCESS PERIODOGRAM

The sequence of times $\{X_n; n = 1, 2, \dots, N\}$ may be regarded as the first N elements of the point process associated with the counting process $\{N_t\}$, which starts at 0 and jumps by 1 at each of the times X_n . The periodogram of $\{N_t; 0 < t \leq T\}$ is defined by [12]

$$I(f) = \frac{2}{T} \left| \int_0^T e^{2\pi j f t} dN(t) \right|^2 = \frac{2}{T} \left| \sum_{n=1}^N e^{2\pi j f X_n} \right|^2$$

where $X_N \leq T < X_{N+1}$.

As the $2/T$ term is arbitrary, we define the periodogram as

$$I_X(f) = \frac{2}{N} \left| \sum_{n=1}^N e^{2\pi j f X_n} \right|^2,$$

to accord with the multiplicative constant in the usual periodogram definition. Note that this is *not* the usual (discrete-time) periodogram of the $\{X_n\}$ and cannot be computed via

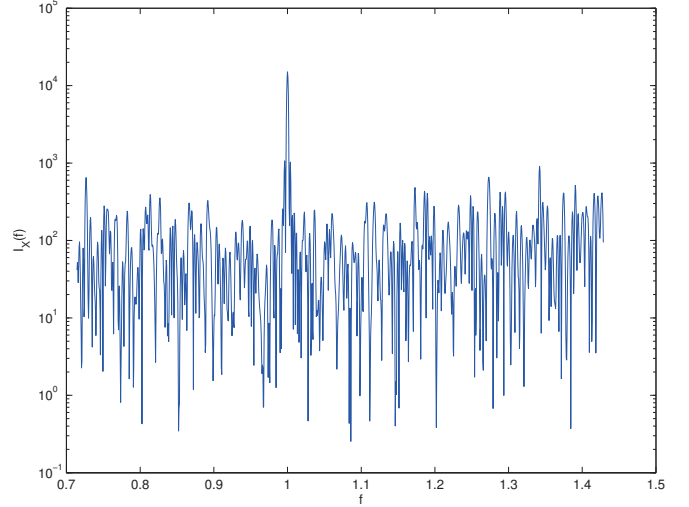


Fig. 1. Point Process Periodogram

FFTs. Instead, it is the (continuous-time) periodogram of a sequence of 1's associated with the times $\{X_n\}$. The estimator of β is given by

$$\hat{\beta}_N = \left\{ \arg \max_{f \in \left[\frac{1}{2\beta_L}, \frac{1}{\beta_L} \right)} I_X(f) \right\}^{-1}. \quad (2)$$

Figure 1 depicts $I_X(f)$ for a sample generated by (1) with $\beta_0 = 1$, $\beta_L = 0.7$, $\sigma_U = 0.05$. A peak may be seen near $f = 1$, and the appearance is similar to that of the usual periodogram of a noisy sinusoid.

A peak in $I_X(f)$ is expected near $f = f_0 = 1/\beta_0$, since

$$\frac{1}{N} \left| \sum_{n=1}^N e^{j2\pi f_0 X_n} \right| = \frac{1}{N} \left| \sum_{n=1}^N e^{j2\pi f_0 \varepsilon_n} \right| \rightarrow \phi_U(2\pi),$$

almost surely as $N \rightarrow \infty$, by the strong law of large numbers. When σ_U^2 is small, $\phi_U(2\pi)$ is close to 1. On the other hand, when f differs from β_0^{-1} , the S_n terms are not cancelled, and $I_X(f)$ tends to be small compared with N . This was noted in [1] and [2].

IV. SIMULATIONS

The following figures summarise the results of simulations for $N = 16, 32, 64, 128$ and four different Y_j distributions. In each case, $\beta_L = 0.7$, $\beta_0 = 1.0$, the U_n were (simulated) zero-mean and Gaussian with $\sigma_U^2 = (0.05)^2 \times 2^{k/3}$, $k = -4, \dots, 18$. The Y_j were (simulated)

- 1) Poisson with mean 1;
- 2) Poisson with mean 3;
- 3) Geometric (starting at 0) with mean 1.
- 4) Geometric (starting at 0) with mean 3.

The periodogram was maximised over $(10N + 1)$ equispaced points between $(2\beta_L)^{-1}$ and $(\beta_L)^{-1}$, and Newton's

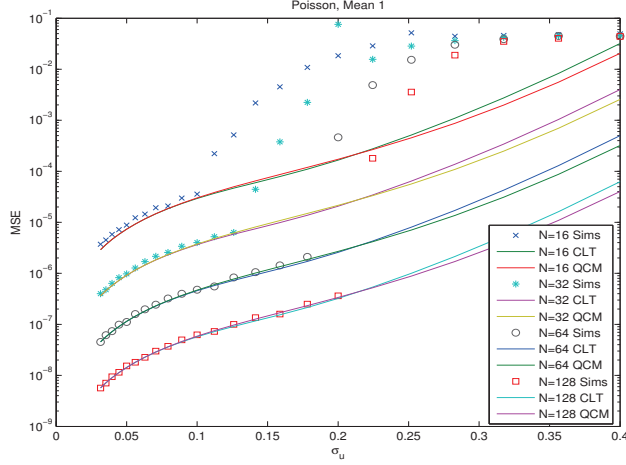


Fig. 2. Simulation results: Poisson(1)

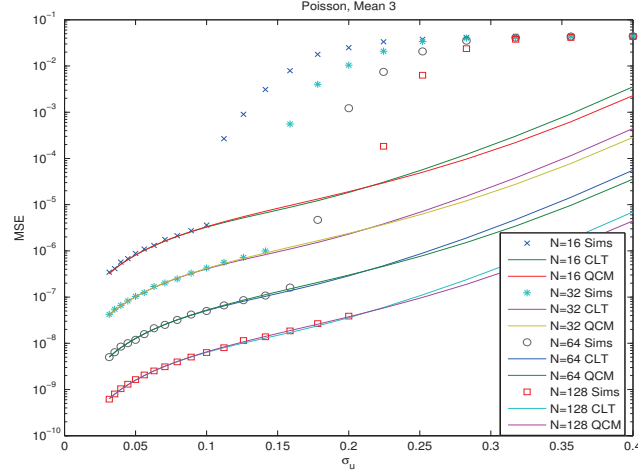


Fig. 3. Simulation results: Poisson(3)

method was then used to find the closest zero of $I'_X(f)$ to the initial maximiser. For each combination of N and σ_U^2 , 2000 replications were undertaken, and the sample mean square errors of the $\hat{\beta}_s$ computed. It is clear from the plots, where CLT and QCM stand for the theoretical asymptotic variances of the periodogram estimator and the modified least squares estimator [9], that the theory and simulations agree as σ_U^2 increases, until a thresholding effect is noticeable, occurring as expected at values of σ_U^2 that depend on N .

V. STATISTICAL PROPERTIES

The following two theorems describe the main results of this paper. They are followed by necessary lemmas and their proofs. Finally, we give the proofs of the theorems.

Theorem 1: $N(\hat{\beta}_N - \beta_0) \rightarrow 0$ almost surely as $N \rightarrow \infty$.

Theorem 2: The distribution of

$$N^{3/2}(\hat{\beta}_N - \beta_0)$$

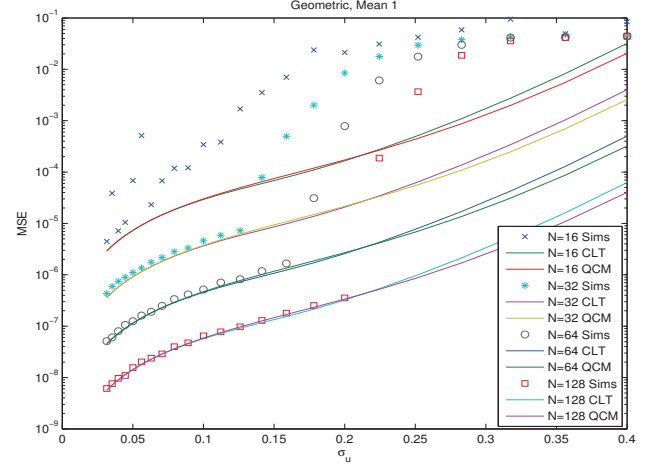


Fig. 4. Simulation results: Geometric, mean 1

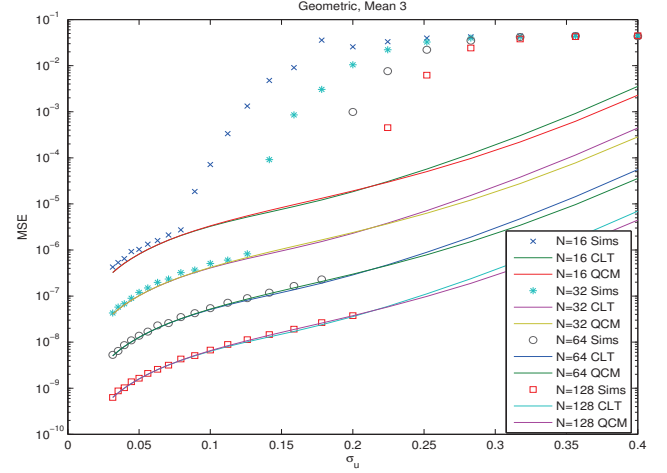


Fig. 5. Simulation results: Geometric, mean 3

converges as $N \rightarrow \infty$ to the normal with mean 0 and variance

$$\frac{3\beta_0^2\{1 - \phi_U(4\pi)\}}{2\pi^2\mu_Y^2\phi_U^2(2\pi)}.$$

The following two lemmas are used to demonstrate the rate at which the periodogram converges, firstly for values on a discrete grid of frequencies, and then for all frequencies. The proofs are omitted, but are complicated by the fact that two sources of randomness, namely $\{\varepsilon_n\}$ and $\{Y_n\}$ are involved.

Lemma 3: Let

$$h_N(\delta) = \frac{1}{N} \sum_{n=1}^N e^{j2\pi f X_n}$$

$$T_N(\delta) = E\{h_N(\delta)\},$$

and

$$W_N(\delta) = h_N(\delta) - T_N(\delta),$$

where $\delta = f\beta_0 - 1$. For some $d > 0$, and $\xi > 0$, let

$$\delta_i = -\frac{1}{2} + \frac{3i}{2[N^{d+1}]}, \quad i = 0, 1, \dots, (1 - \xi)[N^{d+1}].$$

Then

$$\sup_i |W_N(\delta_i)| \rightarrow 0,$$

almost surely as $N \rightarrow \infty$.

Lemma 4:

$$\sup_{\delta} |W_N(\delta)| \rightarrow 0,$$

almost surely as $N \rightarrow \infty$.

The following three lemmas are used to show that the (scaled) periodogram converges uniformly in a neighbourhood of the true frequency to a function that is uniquely maximised at the true frequency. This behaviour is required to show that the periodogram maximiser converges to the true frequency at the desired rate.

Lemma 5: For fixed δ ,

$$\lim_{N \rightarrow \infty} T_N(\delta) = \begin{cases} \kappa & ; \delta = 0 \\ 0 & ; \delta \neq 0, \end{cases}$$

where $\kappa = \exp(j2\pi\alpha_0/\beta_0)\phi_U(2\pi)$.

Lemma 6: For fixed δ ,

$$\begin{aligned} \lim_{N \rightarrow \infty} T_N\left(\frac{\delta}{N}\right) &= \tau(\delta) \\ &= \kappa \begin{cases} 1 & ; \delta = 0 \\ \frac{\exp(j2\pi\delta\mu_Y) - 1}{j2\pi\delta\mu_Y} & ; \delta \neq 0, \end{cases} \end{aligned}$$

which is a continuous function of δ .

Lemma 7: $|\tau(\delta)|^2$ is uniquely maximised at $\delta = 0$.

Proof:

$$\frac{|\tau(\delta)|^2}{\phi_U^2(2\pi)} = \frac{\sin^2(\pi\delta\mu_Y)}{(\pi\delta\mu_Y)^2} \leq 1,$$

with equality if and only if $\delta = 0$. ■

Proof: (Theorem 1) Let $\hat{\delta}_N$ maximise $|h_N(\delta)|^2$ over $[\beta_0/(2\beta_L) - 1, \beta_0/\beta_L - 1]$. Then

$$|h_N(\hat{\delta}_N)|^2 \geq |h_N(0)|^2 \rightarrow |\tau(0)|^2.$$

Since δ is confined within a compact set, for each realisation of $\{\hat{\delta}_n; n = 2, 3, \dots\}$ there is a subsequence that converges. Suppose that for some realisation, there is a subsequence that converges to δ' . Now, lemma 7 shows, since

$$|\tau(\delta')|^2 \geq |\tau(0)|^2,$$

that $\delta' = 0$, and consequently that the only possible limit point of $\{\hat{\delta}_n\}$ is 0. Since the results hold almost surely, it follows that $\hat{\delta}_N$ converges almost surely to 0. In light of this fact and lemma 4, it also follows that $N\hat{\delta}_N$ converges almost surely to 0. The reasoning is similar to that used in [13] and [14]. ■

Proof: (Theorem 2) Put $g_N(\delta) = \frac{1}{2}|h_N(\delta)|^2$. We can omit in what follows the term $\exp(j2\pi\frac{\delta+1}{\beta_0}\alpha_0)$ in the definition of $h_N(\delta)$, since its modulus is 1. Now

$$0 = g'_N(\hat{\delta}_N) = g'_N(0) + g''_N(\tilde{\delta}_N)\hat{\delta}_N,$$

where $\tilde{\delta}_N$ is between $\hat{\delta}_N$ and 0. Thus

$$N^{3/2}\hat{\delta}_N = -\frac{N^{-1/2}g'_N(0)}{N^{-2}g''_N(\tilde{\delta}_N)}.$$

We shall show that $N^{-1/2}g'_N(0)$ is asymptotically normal and that $N^{-2}g''_N(\tilde{\delta}_N)$ converges to a constant. The following computations are needed, and are included here for completeness:

$$g'_N(\delta) = \text{Re}\{h'_N(\delta)h_N^*(\delta)\}$$

$$g''_N(\delta) = |h'_N(\delta)|^2 + \text{Re}\{h''_N(\delta)h_N^*(\delta)\}$$

$$h_N(0) = \frac{1}{N} \sum_{n=1}^N e^{j2\pi U_n}$$

$$h'_N(\delta) = \frac{j2\pi}{N} \sum_{n=1}^N (S_n + U_n) e^{j2\pi U_n} e^{j2\pi\delta\{S_n+U_n\}}$$

$$h'_N(0) = \frac{j2\pi}{N} \sum_{n=1}^N (S_n + U_n) e^{j2\pi U_n}$$

$$h''_N(\delta) = -\frac{(2\pi)^2}{N} \sum_{n=1}^N (S_n + U_n)^2 e^{j2\pi U_n} e^{j2\pi\delta(S_n+U_n)}.$$

Since $N\tilde{\delta}_N \rightarrow 0$, almost surely as $N \rightarrow \infty$, it follows that

$$\frac{g''_N(\tilde{\delta}_N)}{g''_N(0)} \rightarrow 1$$

almost surely as $T \rightarrow \infty$.

Simple calculations show that

$$N^{-2}g''_N(0) \rightarrow -\frac{(2\pi)^2}{12} \mu_Y^2 \phi_U^2(2\pi)$$

in probability as $T \rightarrow \infty$. Lyapunov's central limit theorem [15] may be used to show that $N^{-1/2}g'_N(0)$ is asymptotically normally distributed with mean 0 and variance

$$(2\pi)^2 \mu_Y^2 \phi_U^2(2\pi) \frac{1 - \phi_U(4\pi)}{24}$$

and so $N^{3/2}\hat{\delta}_N$ is asymptotically normally distributed with mean 0 and variance

$$\frac{(2\pi)^2 \mu_Y^2 \phi_U^2(2\pi) \frac{1 - \phi_U(4\pi)}{24}}{\left\{ \frac{(2\pi)^2}{12} \mu_Y^2 \phi_U^2(2\pi) \right\}^2} = \frac{3\{1 - \phi_U(4\pi)\}}{2\pi^2 \mu_Y^2 \phi_U^2(2\pi)}.$$

But

$$N^{3/2}(\hat{\beta} - \beta_0) = -\beta_0 \frac{N^{3/2}\hat{\delta}_N}{1 + \hat{\delta}_N},$$

which is therefore asymptotically normal with mean 0 and variance

$$\frac{3\beta_0^2\{1 - \phi_U(4\pi)\}}{2\pi^2 \mu_Y^2 \phi_U^2(2\pi)},$$

since $1 + \hat{\delta}_N$ converges in probability to 1 as $N \rightarrow \infty$. ■

Note: since the least squares estimator of β when the S_n are known is

$$\hat{\beta}_{LS} = \frac{\sum_{n=1}^N X_n(S_n - T_N)}{\sum_{n=1}^N (S_n - T_N)^2},$$

where

$$\begin{aligned} T_N &= N^{-1} \sum_{n=1}^N S_n \\ &= N^{-1} \sum_{n=1}^N \sum_{i=1}^n Y_i \\ &= N^{-1} \sum_{i=1}^N (N+1-i) Y_i, \end{aligned}$$

the variance of $N^{3/2}(\hat{\beta}_{LS} - \beta_0)$ conditional on $\{Y_1, Y_2, \dots, Y_N\}$, (N^3 multiplied by the “clairvoyant” Cramér-Rao lower bound) is

$$\begin{aligned} &N^3 E \left\{ \frac{\sum_{n=1}^N X_n(S_n - T_N)}{\sum_{n=1}^N (S_n - T_N)^2} - \beta_0 \right\}^2 \\ &= N^3 E \left\{ \frac{\sum_{n=1}^N \varepsilon_n(S_n - T_N)}{\sum_{n=1}^N (S_n - T_N)^2} \right\}^2 \\ &= \frac{N^3 \text{var } \varepsilon_n}{\sum_{n=1}^N (S_n - T_N)^2}. \end{aligned}$$

But

$$\begin{aligned} \sum_{n=1}^N (S_n - T_N)^2 &= \sum_{n=1}^N \left\{ \left(n - \frac{N+1}{2} \right) \mu_Y + O_P(n^{1/2}) \right\}^2 \\ &= \frac{N^3}{12} \mu_Y^2 + O_P(N^{5/2}), \end{aligned}$$

since

$$S_n = n\mu_Y + \sum_{j=1}^n (Y_j - \mu),$$

and

$$\sum_{j=1}^n (Y_j - \mu) = O_P(n^{1/2}),$$

by the central limit theorem. The (exact) variance of $N^{3/2}(\hat{\beta}_{LS} - \beta_0)$ conditional on $\{Y_1, Y_2, \dots, Y_N\}$ is thus

$$\frac{12\beta_0^2\sigma_U^2}{\mu_Y^2} + o_P(1).$$

The ratio of the asymptotic variance of the estimator $\hat{\beta}_N$ to the conditional asymptotic variance of $\hat{\beta}_{LS}$ is therefore

$$\frac{1 - \phi_U(4\pi)}{8\pi^2\sigma_U^2\phi_U^2(2\pi)}.$$

When $\{U_n\}$ is Gaussian, this ratio is

$$\frac{1 - \exp(-8\pi^2\sigma_U^2)}{8\pi^2\sigma_U^2 \exp(-4\pi^2\sigma_U^2)},$$

which is easily shown to be greater than 1, but which is, as σ_U^2 tends to 0,

$$\frac{1 - 1 + 8\pi^2\sigma_U^2 + O(\sigma_U^4)}{8\pi^2\sigma_U^2\{1 + O(\sigma_U^2)\}} = 1 + O(\sigma_U^2).$$

Thus, when the noise variance is “small”, the asymptotic variance of $\hat{\beta}_N$ is close to what it would be if the S_n were known.

VI. CONCLUSION

We have described the theoretical (asymptotic) statistical properties of the Fogel and Gavish “periodogram” estimator of period, and indicated how they are derived. The properties have been illustrated and confirmed by means of a number of simulations. We have also compared the asymptotic properties with those of another least-squares based technique developed by the authors.

REFERENCES

- [1] E. Fogel and M. Gavish, “Parameter estimation of quasi-periodic sequences,” in *Proc. Internat. Conf. Acoust. Speech Signal Process.*, vol. 4, pp. 2348–2351, Apr. 1988.
- [2] E. Fogel and M. Gavish, “Performance evaluation of zero-crossing-based bit synchronizers,” *IEEE Trans. Commun.*, vol. 37, no. 6, pp. 663–665, Jun. 1989.
- [3] S. D. Casey and B. M. Sadler, “Modifications of the Euclidean algorithm for isolating periodicities from a sparse set of noisy measurements,” *IEEE Trans. Signal Process.*, vol. 44, no. 9, pp. 2260–2272, Sept. 1996.
- [4] B. M. Sadler and S. D. Casey, “On periodic pulse interval analysis with outliers and missing observations,” *IEEE Trans. Signal Process.*, vol. 46, no. 11, pp. 2990–3002, Nov. 1998.
- [5] I.V.L. Clarkson, S.D. Howard, and I.M.Y. Mareels, “Estimating the period of a pulse train from a set of sparse, noisy measurements,” in *Proc. Internat. Sympos. Signal Process. Appl.*, vol. 2, pp. 885–888, Aug. 1996.
- [6] N.D. Sidiropoulos, A. Swami, and B.M. Sadler, “Quasi-ML period estimation from incomplete timing data,” *IEEE Trans. Signal Process.*, vol. 53, no. 2, pp. 733–739, Feb. 2005.
- [7] I.V.L. Clarkson, “Approximate maximum likelihood period estimation from sparse, noisy timing data,” *IEEE Trans. Signal Process.*, vol. 56, no. 5, pp. 1779–1787, May 2008.
- [8] R.G. McKilliam and I.V.L. Clarkson, “Maximum likelihood period estimation from sparse, noisy timing data,” in *Proc. Internat. Conf. Acoust. Speech Signal Process.*, pp. 3697–3700, March 2008.
- [9] B.G. Quinn, R.G. McKilliam and I.V.L. Clarkson, “Estimating period from sparse, noisy timing data,” in *Proc. IEEE Statistical Signal Processing Workshop*, pp. 193–196, August 2012.
- [10] S. Dixon, “Automatic extraction of tempo and beat from expressive performances,” *J. New Music Res.*, vol. 30, no. 1, pp. 39–58, March 2001.
- [11] R.G. McKilliam, B.G. Quinn, I.V.L. Clarkson, and B. Moran, “Frequency estimation by phase unwrapping,” *IEEE Trans. Signal Process.*, vol. 58, no. 6, pp. 2953–2963, June 2010.
- [12] M.S. Bartlett, “The Spectral Analysis of Point Processes,” *J. Roy. Statist. Soc. B*, vol. 25, no. 2, pp. 264–296, 1963.
- [13] E.J. Hannan, “The estimation of frequency,” *J. Applied Probability*, vol. 10, pp. 510–519, 1973.
- [14] R.I. Jennrich, “Asymptotic properties of nonlinear least squares estimators,” *Ann. Math. Statist.*, vol. 40, pp. 633–643, 1969.
- [15] P. Billingsley, *Probability and Measure*, Wiley, 1995.